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Fluid Concepts and Creative Analogies: A Theory and its Computer Implementation

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Abstract

We describe a computer model of how humans use concepts fluidly in order to create analogies. Our model is centered on the Slipnet, a network of overlapping concepts whose shapes are determined dynamically by the situations faced by the program. Reciprocally, the state of the Slipnet controls how the program perceives situations. The heart of what the program does, give two situations, is to produce a worlds-mapping: a course-grained mental correspondence between the situations, involving two interdependent and mutually consistent facets: an object-to-object mapping realized in structures called bridges, and a concepts-to-concept mapping realized in structures called pylons. Each pylon expresses a so-called conceptual slippage, borrowed from the Slipnet. Taken together, the slippages constitute a recipe for translating actions in one situation into their analogues in the other. Through the "coattails effect", slippages can induce closely related slippages, allowing deeper and more subtle analogies to be produced than would otherwise be possible.

We believe this model has the potential to create analogies with insight. It has already been realized in one idealized domain, but needs extension. We are continuing our work in that domain, and in addition are trying the model in two new domains, each of which should push our ideas in new directions and at the same time provide a stringent test for them.

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Overview

This article concerns Copycat, Tabletop, and Letter Spirit, three closely related projects in which our research group (the Fluid Analogies Research Group at the University of Michigan) is (1) developing a computational model of creative analogy-making, based on a specific model of "fluid" (i.e., highly flexible) concepts, and (2) testing the model in different domains. The overall goal of these research projects is to explore the nature of human concepts -- their associative structure, their blurry edges, their flexibility as a function of context, and so on -- and we know of no more direct way to probe these phenomena than by the modeling of analogy-making. We believe that creativity is a by-product of having fluid concepts, so that if our model of fluidity "holds water", so to speak, we will automatically have a non-trivial model of creativity as well.

The first major section of this article describes Copycat, our most advanced piece of work. Over the past two years, we have worked on implementing Copycat, and it is now a fully operational (though by no means finalized) computer program, whose probabilistic Slipnet architecture (originally proposed in [Hofs84b]) is the core of all our models. Having described the current state of Copycat, we proceed, in the second major section, to discuss several of the most interesting extensions to Copycat that we plan to carry out. Then we turn to Tabletop and Letter Spirit, describing them in less detail, since they are not yet implemented. Nonetheless, there is plenty of solid material to present, since both projects have been carefully planned out for some time, and the challenges they offer are full of subtlety.

We conclude the article with a brief chronology of our earlier work, and a detailed discussion of how we see our research relating to that of other people working along similar lines in artificial intelligence and cognitive science.

Section I: The Copycat Project -- Where We Are Now

Introducing Copycat and its Microworld: A Simple Analogy

From an examination of many real-life analogies and careful study of the work of other researchers [Evan68, Holy83, Carb83, Gent83, Wins83, Grei85, Keda85, Burs86], we believe we have distilled the recurrent issues of analogy-making [Hofs85a]. To explore those issues efficiently, we have designed a microworld large enough that analogies involving any desired combination of those issues can be constructed in it [Hofs84a]. To start out, let us consider the issues raised by this simple analogy problem:

abc --> abq
ijk --> ?

In the top line, **abc** (the initial string) is transformed into **abq**. Copycat's task is to apply "the same transformation" to **ijk** (the target string).

Letters of the alphabet are narrowly conceived by Copycat: shapes, sounds, words, and all other linguistic and graphic facts are unknown. Copycat knows the alphabet equally well forwards and backwards, but only locally: each letter is directly connected only to its immediate predecessor and successor. Ordinal positions in the alphabet (e.g., **Q** is letter #17) are not known. (Note on notation: Boldface capitals, denoting the 26 abstract categories of the alphabet, never appear in strings; boldface smalls, denoting instances of those categories, appear only in strings.) **A** and **Z**, being extremities, are landmarks of equal importance. **B** can be identified as "the successor of **A**", thus locating it with reference to a landmark. Likewise, **C** can be identified as "the successor of the successor of **A**". However, as one recedes from landmarks, such descriptions grow in length and therefore become less and less probable. Finally, repetitive groups (e.g., **sssss**), successor groups (e.g., **abc**), and predecessor groups (e.g., **kji**) are readily perceived by the program, which is sensitive to sameness and alphabetic closeness, especially between adjacent letters.

In the problem above, two reasonable responses to "What changed?" are: (1) the rightmost letter, or (2) an instance of **C**. The former seems more abstract, and people tend to prefer it. We have accordingly biased Copycat in the same way. How was it changed? Since Copycat cannot formulate any relationship between the initial string and the new letter (the letters in **abc** are too far from **Q**), it sees that letter merely as "a **Q**", and thus answers **ijq**, applying the rule *Replace the rightmost letter by a Q.* However, Copycat might occasionally answer **ijk**, creating a different rule to explain the change, namely, *Replace any C by a Q.*

Variant Analogies Reveal the Interaction of Many Pressures

Here is a nontrivial variant of our initial problem:

abc --> abd
ijk --> ?

Unlike **Q**, the new letter can be "justified" in this context: **D** is the successor of **C**. Copycat thus has several ways of describing what happened, such as:

(1) Replace the rightmost letter by its successor;
(2) Replace the rightmost letter by a **D**;
(3) Replace any **C** by a **D**.

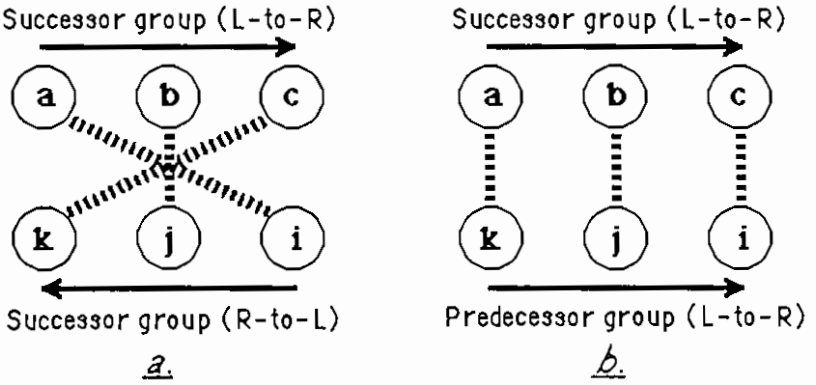
Most humans (and Copycat) favor the more abstract rule (1), producing **ijl**.

Here is a variant in which the target string has been reversed:

abc --> abq
kji --> ?

We could apply the originally favored rule: *Replace the rightmost letter by a Q*, producing **kjq**. But Figure 1a suggests another possibility:

Figure 1.



In it, **abc** is depicted as a normal successor group, and **kji** as a backwards successor group, with the **K** playing the role of the **C**. This reversed mapping justifies answer **qji**. By contrast, **kjq** is based on the direct mapping shown in Figure 1b.

A curious thing occurs when we combine the preceding two variants:

abc --> abd
kji --> ?

The reversed mapping still applies, producing **lji**. But now the direct mapping offers a new

perspective: **abc** is a successor group and **kji** is a predecessor group, suggesting that in the downward leap, the concepts "successor" and "predecessor" switch roles. Moreover, the concept "successor" is an ingredient in the rule: *Replace the rightmost letter by its successor*. These observations strongly suggest the answer **kjh**, in which the rightmost letter is replaced by its predecessor.

Which answer Copycat produces depends on the resemblances on which it grounds its mapping of the two situations. Preserving spatial direction entails a role-switch of concepts "predecessor" and "successor", whereas preserving alphabetic order forces a directional reversal. Both perspectives on **kji** provide strong answers to this puzzle.

Any real-world situation has innumerable facets, many of which compete as grounds for resemblance to some other situation. Fortunately, the human brain is designed to allow myriad fighting pressures to be resolved. The Copycat architecture has likewise been designed to allow many pressures to coexist, influence each other, and jointly determine an outcome. Our puzzle's variants illustrate how we can manipulate pressures in an analogy. Among the variables we can control are:

(1) the category membership of an object (letter, group, group of groups, etc);
(2) the roles an object plays;
(3) how salient an object is;
(4) how distant an object is from landmarks;
(5) how salient a category is;
(6) how easy it is to perceive groups;
(7) the level of abstractness of an object's description;
(8) the degree of similarity between two objects or roles.

This means that in the microworld, there are hundreds of variants of our simple puzzle, each a unique constellation of these fundamental pressures.

Overview of Copycat's Architecture

The fact that on different runs, Copycat can produce different answers reveals that it is not deterministic. In fact, Copycat's low-level workings are permeated by probabilistic choices. Copycat has no top-level executive; rather, its processing is carried out by small tasks selected probabilistically from a large pool. Each task, when created, is assigned an urgency, based on many factors, determining its probability of being selected.

For Copycat, the main steps in creating an analogy are: (1) perceiving and describing the strings; (2) constructing a rule characterizing the transformation of the inital string; (3) making a mapping between the initial and target strings; and (4) using the mapping to fluidly adapt the rule to fit the target string. (This paper will describe in detail only steps (3) and (4); see also [Hofs84b].) These goals are achieved by many interleaved chains of small tasks, each working on a specific job. In the problem above, one chain might detect that **abc** is a successor group, while another determines that the C is the object that changed, followed by another that finds the object in **ijk** corresponding to the C. Many chains -- some cooperating, some competing -- progress in parallel at different rates, the rate of each being set by the urgencies of its component tasks. Thus Copycat follows the principle of the parallel terraced scan: simultaneously exploring promising avenues at high speeds, and others at lower speeds [Hofs83, Hofs84b]. Roughly speaking, good answers can be reached via many different high-probability pathways and poor answers via few pathways of low probability. This way, Copycat's low-level stochastic nature manifests itself macroscopically as Copycat's "taste".

Copycat's architecture has two interdependent components: the tasks described above, and a network of concept nodes and dynamic "conceptual distances" called the Slipnet [Hofs79, Quil68]. This structure, attempting to capture our notion of "concept", is the heart of our model.

The Slipnet

Figure 2 shows a small portion of Copycat's Slipnet. A node represents a concept (more precisely, the core of a concept), and outgoing sliplinks establish proximities to other concepts.

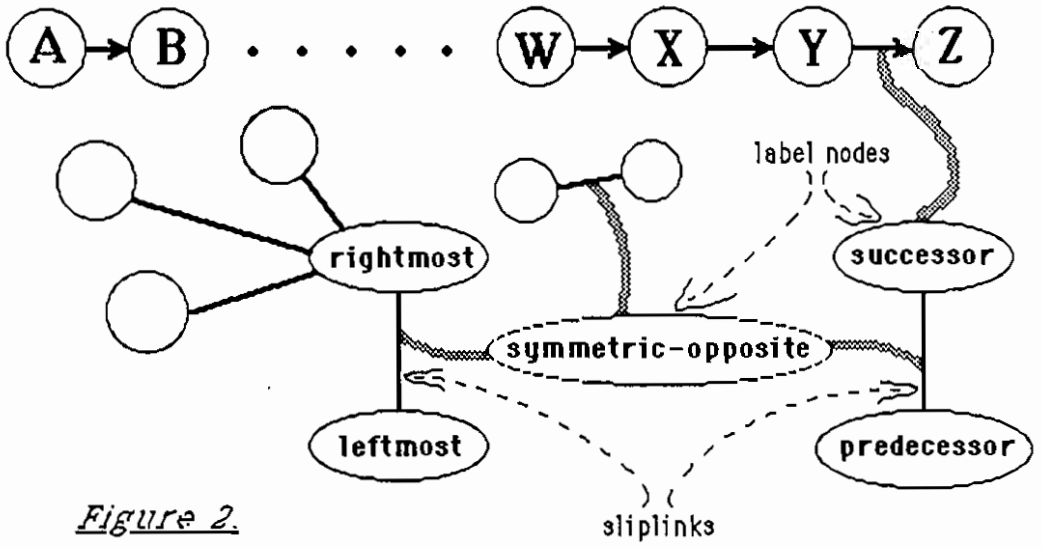


Figure 2.

Each node has a fixed semanticity (not shown): a number representing the abstractness or generality of the concept involved. Letters have low semanticities; concepts like "predecessor" and "right-neighbor" are more semantic; then come concepts like "repetitive group"; finally, domain-independent concepts like "symmetric-opposite" have the highest semanticities of all.

Each node also has a time-varying activation [Reit65, Norm82] -- a number representing its perceived relevance. Slipnet nodes become activated when the system perceives (via chains of tasks) instances of the concept involved, and stay activated as long as such instances are perceived. The activation of a node pushes the system to look for more instances of the given concept. Nodes spread activation to their neighbors [Coll75], and also decay [Ande83]. Low-semanticity nodes lose activation quickly through decay, hence never get very activated, whereas high-semanticity nodes, decaying slowly, can reach high activations. This ensures that abstract concepts have deeper influences on the course of the analogy [Gent83].

A sliplink simultaneously represents a resemblance of two concepts and a potential slippage of either concept into the other. Sliplinks have two attributes: length and label. Length corresponds to the intuitive notion of conceptual distance. A link's length changes dynamically, in a manner determined by its label. Thanks to such distortions, concepts change shape over time (see below). A sliplink's label roughly classifies the potential slippage the link encodes. For example, "symmetric-opposite" labels both the sliplink joining "rightmost" and "leftmost" and the sliplink joining "predecessor" and "successor". By contrast, the sliplink from **P** to **Q** is labeled "successor". Also, from any node to itself there is a virtual sliplink of constant length zero, labeled "identity".

Sliplinks' labels are themselves concepts, and so are represented by Slipnet nodes. The activation of such a node determines the lengths of all sliplinks it labels: as it grows more active, those sliplinks shrink; as it loses activation, they grow longer. Thus, a local phenomenon (the changing activation of one node) has global repercussions (the changing shape of many concepts). An important aspect of Copycat is this interdependence of perception and concepts: what Copycat "sees" affects (through activation) the shapes of its concepts, which in turn affect (through pressure on the tasks doing the looking) how Copycat perceives its world.

Here is how, in our view, concepts are embodied. Crudely, a concept is a region in the Slipnet, centered on a particular node (its core), often including nearby nodes, and whose shape can vary with time. But this implies that a concept has sharp boundaries, so that a given node either is or is not inside it, whereas everyday life and experiment [Smit81, Hofs85a, Holl86] tell us that concept membership is blurry. Consider the concept "food". Obviously, spaghetti is a food. Is ketchup? The cloud fades near cough syrup and chewing gum, and only in an emergency would we include grass and shoes. Undeniably outside the "food" cloud are Anglican cathedrals, the Spanish Inquisition, and the moon's orbit. Accordingly, our definition of "concept" is: a region in the Slipnet whose density peaks at its core, and which tails off with conceptual distance from the core. Since sliplink lengths are variable, a concept's contours change over time. We emphasize that concepts are not explicit structures that can be manipulated by Copycat; rather, they are emergent aspects of the Slipnet in terms of which activity can be intuitively envisioned by humans. This notion of emergent concepts [Hofs85b] arising from many low-level interactions between dynamic link-lengths and environmental inputs gives Copycat's Slipnet a flavor similar to many connectionist models [McCl86, Kahn86].

Abstraction and Robust Rules: A More Complex Analogy

An important principle of analogy-making is that once a coarse-grained mapping between two situations has been set up, involving only their most salient aspects, the rest of the analogy follows [Gick83]. We call such a mapping a worlds-mapping. Below, we show how Copycat attempts to see the essence of a complex situation by formulating a robust rule, and show how it creates a worlds-mapping allowing it to export that rule to a new context. Here is the problem:

ssskkkooooooooo --> ssskkkopooooooo
oooooookkss --> ?

A crude rule for this transformation is: *Replace the eighth letter by a P,* yielding **ooooooopkss**. We dismiss this as being too "brittle". Brittle rules are literal and myopic and take few structural features into account, whereas robust rules characterize places and categories more abstractly, taking into account the structure of the situations [Gent83].

In Copycat, a competition occurs for constructing a robust rule. Early on, several descriptions are constructed for each object; one of these must be chosen if the object figures in the rule. Each description has a "desirability" value, which favors highly semantic and/or highly active concepts, and discriminates against verbosity. In constructing a rule, Copycat probabilistically selects one description for each object involved, favoring high-desirability descriptions.

Accordingly, Copycat will most likely produce this rule: *Replace the right-neighbor of the leftmost letter in the rightmost group by its successor,* which seems abstract and robust. Unfortunately, when applied to the target string **oooooookkss**, it gives an answer, **oooooookkst**, that takes no account of the strong (though inexact) resemblance between the initial string written in reverse order and the target string. Most humans, rather than changing a letter in the target's rightmost group, **ss**, would change one in its leftmost group, **ooooooo**. For such flexibility to arise in Copycat, the worlds-mapping must interact with and translate the robust rule.

How a Worlds-Mapping Induces Rule Translation

Since no rule can anticipate how all other worlds will differ from the original one, even the most robust of rules will have to be translated. (Rule translation is closely related to prototype deformation [Srid81].) In Copycat (and, we think, in human analogy-making), the worlds-mapping dictates the translation. A worlds-mapping consists of two interdependent facets: (1) an object-to-object mapping realized in bridges, and (2) a concept-to-concept mapping realized in pylons. The pylons embody translation rules called slippages. Let us see how the worlds-mapping shown in Figure 3 interacts with the rule to produce a deeper answer (how Copycat constructs such a worlds-mapping is explained subsequently).

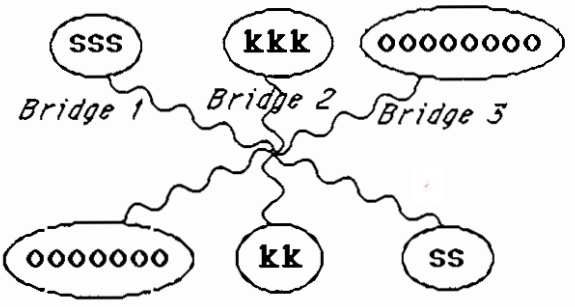


Figure 3.

This mapping can interact with the rule in two interesting ways, one rigid and one fluid. The first way yields **opoooookkss**. Underlying this is the following translation:

Raw rule:
Replace the right-neighbor of the leftmost letter in the rightmost group by its successor.
|
V
Translated rule:
Replace the right-neighbor of the leftmost letter in the leftmost group by its successor.

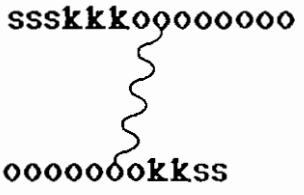
This translation, derived from Bridge 3, appears to be a substitution of a description of its lower terminus for a description of its upper terminus. In reality, however, the relation to Bridge 3 is subtler, as we explain below.

What about the more fluid approach, yielding **ooooopokkss**? Underlying this would appear to be the following translation:

Raw rule:
Replace the right-neighbor of the leftmost letter in the rightmost group by its successor.
|
V
Translated rule:
Replace the left-neighbor of the rightmost letter in the leftmost group by its successor.

If this substitution was derived from a bridge connecting the two objects in question (as seemed to have happened above), then that bridge could only be the one shown in Figure 4.

Figure 4.



Unfortunately, to have constructed this bridge, Copycat would have had to recognize that those particular O's are counterparts -- but recognizing that is the whole problem!

Resemblances, Bridges, and Pylons

There is a way to avoid circularity, however: we can think of the translated rule as being derived from the raw rule not via this hypothetical bridge which pinpoints the counterpart of the changed letter, but via the coarser-grained, more global bridges of the worlds-mapping (Figure 3). And the desired translation, instead of coming from one single substitution of a big long object description for another object description (as above), is achieved by three separate substitutions of

one concept for another:

Raw rule:
Replace the right-neighbor of the leftmost letter in the rightmost group by its successor.

Translated rule:
Replace the left-neighbor of the rightmost letter in the leftmost group by its successor.

This brings us to two absolutely central premises of Copycat:

(1) The translation step consists of concept substitutions, not object substitutions;
(2) The concept substitutions are derived from pylons supporting bridges.

What makes a bridge? A bridge is supposed to join "counterparts". But what makes two things be counterparts? The answer is resemblances, on concrete or abstract levels. In Copycat, a resemblance between two objects is strong to the degree that the concepts in their descriptions are close in the Slipnet. The context-dependent nature of resemblance is thus modeled by the context-sensitivity of conceptual distances in the Slipnet. Resemblances can be intrinsic (shared or similar internal properties) or extrinsic (shared or similar roles and relationships) [Hofs85a, Burs86]. For instance, in the puzzle, sss and ss have both intrinsic and extrinsic resemblances: they are composed of the same letter, and play similar roles ("leftmost group" and "rightmost group"). Extrinsic resemblances tend to involve concepts of higher semanticity, so they often carry more weight. For this reason, Copycat often favors "systematicity" in analogies [Gent83].

Each resemblance (concept-concept closeness in the Slipnet) that justifies a given bridge is embodied in a concrete data structure: a pylon. Every pylon (real slippage) is thus a copy of a sliplink (potential slippage). The more pylons a bridge has (and the stronger they are), the stronger it is.

Salience, Competition, and the Emergence of a Worlds-Mapping

The Slipnet, being the measure of all resemblances, implicitly decides whether any suggested bridge ought to be built. But what suggests a bridge in the first place? Copycat picks pairs of objects probabilistically, with a strong bias towards salient objects. Such a pair, constituting a suggested bridge, is called a proto-bridge. Our criteria for salience are simple: groups are salient, end-points of strings and groups are salient, instances of salient categories (e.g., **A** and **Z**) are salient, and so on.

The sss in the initial string is salient, being both a group and an endpoint (leftmost group). Similarly, the ss in the target is salient. A proto-bridge between them is thus likely to be suggested, triggering a test to determine if it deserves upgrading to full bridgehood. Three reasons for considering these objects counterparts (and thus building the bridge) are:

(1) both objects are repetitive groups;
(2) both objects are composed of S's;
(3) one is a leftmost object while the other is a rightmost object.

These resemblances correspond to the following slippages embodied in three separate pylons: "repetitive-group ==> repetitive-group", "S ==> S", and "leftmost ==> rightmost". As the bridge from sss to ss is traversed, the first concept in each pylon slips into the second. Of course, the first two are "trivial" slippages (identity slippages), while the last one is non-trivial.

While one bridge is being suggested, evaluated, and built, similar activities can be going on elsewhere. Some proto-bridges have too little support to become full bridges; others, though strong, are incompatible with bridges already in place. For example, an sss-oooooo proto-bridge would contradict an already-existing sss-ss bridge, since sss can correspond to only

one object in the target string. (The general problem of determining if bridges are compatible -- express the same worldview -- is too complex to be dealt with in depth here, but it seems intuitively clear that bridges are compatible if their pylons express conceptually parallel slippages; for instance, slippages "leftmost ==> leftmost" and "rightmost ==> rightmost" are parallel, whereas "leftmost ==> leftmost" and "left-neighbor ==> right-neighbor" are not.) In cases of incompatibility, fights take place whose outcome is probabilistically based on the comparative strengths of the opposing bridges. If the challenger wins, it is erected while the prior bridge is torn down; otherwise, the *status quo* is preserved. From this probabilistic struggle for survival [Hofs83, Kirk83, Hofs84b, Smol84], there evolves a strong, mutually compatible family of bridges, which, together with their pylons, constitute the desired worlds-mapping.

Creativity and the Coattails Effect

The only non-trivial pylons supporting the bridges in the worlds-mapping in Figure 3 are "leftmost ==> rightmost" and "rightmost ==> leftmost", and they yield the following translation:

Raw rule:
Replace the right-neighbor of the leftmost letter in the rightmost group by its successor.

Translated rule:
Replace the right-neighbor of the rightmost letter in the leftmost group by its successor.

This rule, however, gives rise to a very bizarre answer: **ooooooolkss**. To obtain the desired answer, **ooooopokkss**, one further non-trivial slippage, "right-neighbor ==> left-neighbor", is needed. Fortunately, Copycat can discover and carry out this third slippage thanks to the coattails effect, whereby slippages that "win" also bring along sufficiently close relatives.

Consequences of abusing the coattails effect are shown by this joke:

American: Look how free we are -- we can parade before the White House and yell "Down with Reagan!" with no fear of being arrested.
Russian: Big deal -- we too can parade before the Kremlin and yell "Down with Reagan!" without fear.

In the Russian's mind, the "United States ==> Soviet Union" slippage induced the parallel slippage "White House ==> Kremlin", but these slippages did not bring along the very close parallel slippage "Reagan ==> Gorbachev". On a serious level, what this joke illustrates is the importance of consistently carrying through a set of slippages, which is what the coattails effect does.

Conscious usage of the coattails effect to extend an analogy can have powerful consequences. We have observed that many scientific discoveries originate in daring but careful coattails-like extensions. For example, when the nuclear force was first recognized, Hideki Yukawa conceived of a mediating particle for it by analogy to the electromagnetic force, which was known to have a mediating particle (the uncharged, massless photon). However, the new particle would have to mediate conversion of uncharged particles (neutrons) into charged particles (protons) and vice versa, whereas photons could not do so. On the coattails of this "non-convertor ==> convertor" slippage came the parallel slippage "uncharged ==> charged" (i.e., the new particle would itself have to be charged). But closely associated with the concept of being charged was that of being massive (not by ironclad law but by previous examples), and so this second slippage brought along a third parallel slippage: "massless ==> massive". And, since massive particles obey a different equation from massless ones, a fourth parallel slippage came along: "massless equation ==> massive equation". Yukawa then used this equation to predict properties of the new particle (now known as the pion); the particle was discovered roughly a decade later, and those properties were verified [Yuka73]. Because we feel the coattails effect plays a major role in human creativity, it is one of the most important properties of analogy-making we have sought to model.

Intuitively, the worlds-mapping in Figure 3 should cause concepts closely associated with "left" and "right" to switch wherever they occur. The Slipnet proximity of "right-neighbor" to "rightmost" (which we know slips to "leftmost") is sufficient to induce it to slip to "left-neighbor" -- even though this slippage is not expressed by any pylon. Therefore, the actual translation of the rule would consist of three parallel non-trivial slippages:

Raw rule:
Replace the right-neighbor of the leftmost letter in the rightmost group by its successor.

Translated rule:
Replace the left-neighbor of the rightmost letter in the leftmost group by its successor.

A subtle coordination of bridges, pylons, slippages, and the coattails effect thus gives rise to the desired "fluid" answer, **oooooopokkss**. This example shows how Copycat handles many complex aspects of analogy-making.

Section II: Where We Are Headed

In this section, we cover all three projects, beginning with new directions for Copycat, and moving on to descriptions of Tabletop and Letter Spirit. The new projects are compared with Copycat and with each other, in an attempt to bring out the diversity and complementarity, as well as the overlap, of all our projects.

Copycat

Now that Copycat's basic architecture is in place and the program is solving a variety of analogy problems, an immediate and important goal is to make it more sophisticated and flexible. This subsection describes some of our plans for future work on the Copycat project. By no means is it a complete list; we have limited ourselves to discussing a handful of the most interesting challenges. First, we discuss a family of experiments that we wish to carry out on the program as it develops, to see how modification of certain parameters will alter its observable "taste". Second, we describe four concrete ways in which we can improve Copycat's performance without turning it into a hopelessly specialized expert in its alphabetic microworld.

Experiments with Parameter Settings

The Copycat program contains many numerical parameters, all of which are assigned values by us. These parameters include: the semanticities and decay rates of all Slipnet nodes, the amount of activation pumped into Slipnet nodes, the initial lengths of all sliplinks, the frequency with which activation is spread in the Slipnet, the urgencies of various types of codelets, and so on. In addition, other numerical parameters are "hidden" in various formulas used to calculate values such as: the salience of objects, the degree of resemblance between two objects, the strength of bridges, the degree of parallelism of slippages, and so on. One of our major tasks will be to study the effects of different settings for the many parameters in the system.

We proposed above that "cognitive style" or "taste" emerges from the way in which many pressures of all different degrees of importance coexist, cooperate, and compete -- that is, from the probabilistically-governed parallelism of many different chains of tasks of various urgencies, under the influence of numerous global biases, both dynamic (e.g., the activations of concepts) and static (e.g., the semanticities of concepts). It will therefore be of great interest to observe the effects on the system's behavior created by varying parameter settings, and to determine if specific types of behavior (e.g., highly rigid vs. highly fluid) can be elicited by specific parameter-settings.

Making Copycat More Sophisticated

Our continuing goal for Copycat is, of course, for it to be able to solve ever-harder problems in its alphabetic microworld. We need to improve many features of the program to enable it to come anywhere close to solving the full range of problems given in [Hofs84a]. (The Appendix to this article contains a representative sample of those problems.) Below, we describe four specific aspects of the Copycat architecture on which we plan to focus: more flexible rule-construction, more sophisticated perception, paradigm shifts, and temperature-controlled parallelism. All of these are concerned with general aspects of analogy-making, and have very little dependence on specifics of Copycat's microworld.

1. More Flexible Rule-Construction

Currently, Copycat has only one "rule template" [Hofs84b] in which to frame its description of the transformation of the initial string: *Replace* _____ *by* _____ , which is adequate for a situation where exactly one object is to be replaced. Even though this apparently simple rule template actually suffices for some quite complex analogies (such as the examples discussed above), the class of problems that can be solved is fairly limited. For example, Copycat cannot currently solve problems like these two:

axcxe --> abcde
ixkxm --> ?

abc --> aabbcc
ijkl --> ?

The rule in the first problem would be something like *Replace each X by the letter alphabetically between its left-neighbor and its right-neighbor,* and the rule in the second problem would be something like *Replace each letter by two instances of the same letter.* Both rules involve replacing more than one object. Moreover, there are many other classes of analogy problems in the microworld requiring more sophisticated rules [Hofs84a]. Therefore, to explore further issues in human analogy-making, we need to give our program more templates, and a rudimentary ability to combine templates, which will allow it to construct more sophisticated rules.

2. More Sophisticated Perception

The first step towards more complex rule-construction is to enrich Copycat's set of concepts, and to enable it to detect subtler kinds of structures and to formulate more abstract descriptions of objects and their relations to other objects. In the first example above, the program should be able to determine that the **X**'s are "place-holders", and to describe them as such. This entails being able to recognize that **axcxe** is a successor group in disguise.

Also, consider the following problem:

rmj --> jmr
xkpe --> ?

To solve this problem requires noticing that **jmr** is the reverse of **rmj**. Another notion we would like the program to have is that of symmetry, as in this problem:

abc --> abbcccbba
ijkl --> ?

We would like the program to be able to detect the symmetry of **abbcccbba**, and to include that feature in that object's list of descriptions.

An even more central perceptual ability of Copycat is "chunking" or "grouping". Copycat's Slipnet contains the concepts "repetitive group", "successor group", and "predecessor group", and

the program is now able to perceive strings like **abcppp** as being made up of such groups. An extension to Copycat's ability to perceive groups would enable it to solve problems like this one:

rr --> **rrr**
gamgamgam --> **?**

To solve this requires construction of a group of otherwise unrelated elements (**gam**).

A very tricky feat of chunking is required for the following problem:

abc --> **abd**
aababc --> **?**

Firstly, we want the system to see the target string as consisting of three successive chunks: **a**, **ab**, and **abc**. To do this, the system has to be able to overcome the strong "temptation" to chunk the two initial **A**'s into a repetitive group **aa**, by recognizing that the sacrifice of that piece of local order is more than compensated for by the achievement of global order. Secondly (though not necessarily in a chronological sense), Copycat should recognize that the three chunks themselves constitute a rather abstract kind of successor group, and that the "successor", in that abstract sense, of the rightmost element would be **abcd**. This would yield as answer the string **a ab abcd** (here chunked for clarity). Enabling Copycat to create such an insightful and abstract analogy will be challenging, but it is exactly this kind of deep perception, this seeing to the essence of a complex situation, that we wish to model.

These few examples give an idea of the kind of perceptual issues we plan to deal with. The analogy problems in the Appendix illustrate several other types of perceptual issues that will pose difficult challenges for our model.

3. Paradigm Shifts in the Microworld

At present, Copycat cannot deal satisfactorily with the following problem, which we consider central to our research:

abc --> **abd**
xyz --> **?**

(See [Hofs84b, Hofs85a] for extended discussions of this problem.) Currently, Copycat either produces the very rigid answer **xyd**, or -- much more commonly -- runs into a snag and stops. The snag occurs as follows: Copycat formulates the rule *Replace the rightmost letter by its successor*, and creates a direct worlds-mapping between **abc** and **xyz** (i.e., maps the **A** onto the **X**, etc.). Under this worlds-mapping, all slippages are trivial, and so the translated rule is identical to the raw rule. In trying to apply that rule, however, Copycat fails, since **Z** has no successor. It is vital that Copycat be able to recover from such a snag by (step 1) trying to pinpoint the snag's sources; (step 2) changing various pressures as a result; and then (step 3) trying again in the new "mindset".

In this example, Copycat should ideally recover as follows. Since the snag can be localized to the notion "successor of **Z**", Copycat should focus on Slipnet nodes "successor" and **Z** (step 1) by pumping a big jolt of extra activation into them. This will significantly alter the perceptual biases (step 2), so that when Copycat returns to the bridge-building phase (step 3), it is now much more likely to notice the intrinsic resemblance of the **Z** and the **A**. (This resemblance is grounded in the following facts: **A** and **Z** are the initial and final letters of the alphabet, and "initial" and "final" are Slipnet nodes connected by a "symmetric-opposite" link.) This crucial observation activates the highly semantic node "symmetric-opposite", further biasing the perceptual activities, so that Copycat will be much more likely to spot the extrinsic resemblance of the **Z** and the **A** ("rightmost" and "leftmost"). This will enhance the bias favoring "symmetric-opposite"

perceptions. The **C** ("rightmost") will now be seen as closely resembling the **X** ("leftmost"). Construction of **A-Z** and **C-X** bridges will cause Copycat to change its view of **xyz** from "left-to-right successor group" to "right-to-left predecessor group".

In this completely revamped worlds-mapping, the following non-trivial slippages will be expressed in pylons:

first <==> last;
rightmost <==> leftmost;
successor <==> predecessor.

And the rule will therefore be translated as follows:

Raw rule:
Replace the rightmost letter by its successor.
Translated rule:
Replace the leftmost letter by its predecessor.

We consider the resulting answer, namely **wyz**, to epitomize our notion of "fluid analogies". It represents for us the idea of a "miniature paradigm-shift" -- an unexpected and deep slippage (or, perhaps more often, a bundle of parallel slippages) brought about by the revelation of a contradiction or impossibility of some sort. Such radical shifts in point of view do not come out of nowhere, but rather, are the outcome of sufficiently intense pressures. This type of deep shift, we feel, is close to the roots of human creativity.

Answer **wyz** should be contrasted with other possible answers, such as **xyd** (mentioned earlier), **dyz** (rule: *Replace the leftmost letter by a* ***D***), and **yyz** (rule: *Replace the leftmost letter by its successor*). The latter two represent especially interesting varieties of "wrongness", since they mix fluidity with rigidity in different ways. One other answer often suggested by people is **xya**; it is a good answer but violates the linearity of Copycat's alphabet. Most people have already been exposed to circular alphabets, and thus this answer requires less creativity on their part than it would on the part of Copycat. To come up with **xya**, Copycat would have to invent circularity from scratch, which would be wonderful but is beyond our current dreams.

4. Temperature-Controlled Parallelism

A major factor involved in coordinating the recovery from snags (as we foresee it) is computational temperature [Hofs83, Kirk83, Smol84, McC186]. Temperature is a number roughly indicating how close the system "feels" it is to a good answer. For such a measure to exist, the system must have a way of observing its own progress, and estimating the quality of what it has achieved. Copycat presently has no such measure, but we plan to add it. High temperature would indicate disorganization and dissatisfaction with the current state of the world. Temperature would thus be very high at the start of any analogy problem, and would fall as the program builds structures that bring order into its view of the situations involved. Temperature would be lowest when the problem has been solved.

High temperature amounts to a pressure on the system to try to formulate descriptions of objects and to build organizing structures like groups and bridges; previously built structures are easily broken at high temperature, reflecting the idea that the program is not satisfied with them. Low temperature, by contrast, tends to protect previously built structures, reflecting the program's satisfaction with its current view of the world. Whenever a snag is hit, temperature rises sharply, pressuring the system to restructure its view of the situation.

New Projects: Tabletop and Letter Spirit

The second way in which we hope to extend the ideas in Copycat is by applying the probabilistic Slipnet architecture to other carefully designed domains, and seeing what remains invariant and what must slip in order to give good performance. By such triangulation, we hope to reveal the essence of our architecture, as distinguished from more incidental aspects due to particularities of a single domain. At present, the most clearly-planned new domain is Tabletop; Letter Spirit, though well developed in certain respects, remains a more ambitious challenge. We describe them both below.

Tabletop

The Setup and the "Do this!" Game

The Tabletop domain consists of a typical two-person coffee-house table at which two individuals, Henry and Eliza, are seated. On the table are arrayed various items from the following list (any subset is permitted):

- utensils (knife, fork, spoon);
- large and small plates;
- drinking glasses;
- coffee cups;
- napkins;
- ashtrays;
- dispensers (salt and pepper shakers, sugar bowl, napkin dispenser).

It would be fair to say that under normal circumstances, the main defining property of any Tabletop object would be its category membership, and the qualities that define any category are, naturally enough, defined by the Slipnet (to be described below). But Tabletop, like Copycat, understands the categories in its domain in a rather limited way. For instance, all objects are considered to have no spatial extent (i.e., to be point objects). Consequently, they have very few internal properties (e.g., shape, orientation, and color are nonexistent, and size -- except for plates -- is absent). One internal property, however, is that of being either personal or shared: certain types of objects (utensils, plates, cups, etc.) are personal (i.e., "belong" to either Henry or Eliza), while others (e.g., dispensers) are shared. And no matter where it is found, a personal object remains the property of its original owner, and a shared object remains shared. An internal property of cups and glasses is that they may contain various beverages (e.g., tea, coffee, juice, milk), and may be full, empty, or anywhere in between. Finally, of course, each object has a location on the table. The list of possible object types and properties is by no means cast in concrete; we will undoubtedly find ourselves toying with a number of variations, but for now this is what we have in mind.

The game consists of Henry touching (or conceivably moving) some object on the table and saying to Eliza "Do this!" The rules of the game require her to do something. Needless to say, a human will play the role of Henry, while the Tabletop program will assume the role of Eliza.

The Four Key Pressures and the Slipnet

Possession, conceptual proximity, internal properties, and physical proximity are the key pressures that may enter into Eliza's decision about which object to touch (or to move). Possession, as was stated above, is permanent and position-independent. Of the four notions, it is the most straightforward, although it is by no means trivial.

Suppose the table is clear except for two empty coffee cups, one belonging to Henry and one to Eliza. Henry touches his cup and says, "Do this!" Should Eliza touch Henry's cup, following him in a literal-minded fashion, or should she touch her own cup? This is a fairly clear-cut case for

people, who generally prefer symmetric visions to literal-mindedness; in Eliza's place, they would pick up their own cup.

However, a mild variation of this situation pits conceptual proximity against possession. Replace Eliza's cup by a fork. For her to touch her fork in response to Henry's touching his cup seems quite unnatural, because in people's mind's, forks tend to be conceptually very distant from cups, on account of their very different functions. Therefore Eliza's best option is to touch Henry's cup -- a response that, in the preceding situation, would have seemed very literal-minded.

We can tip the balance either way by altering the category of the object owned by Eliza. The two pressures seem roughly to cancel each other out if we give Eliza a glass. When Henry touches his cup and says "Do this!", it seems about equally plausible for Eliza to touch her own glass or Henry's cup. Fine tuning of the pressures is available by twiddling the "full-empty" knob (i.e., adjusting internal properties); for instance, what if Henry's cup is empty whereas Eliza's glass is full of juice? What if Henry's cup is half full of coffee? Or has just a trace of juice in it?

Conceptual resemblances will be represented, of course, by sliplinks in Tabletop's Slipnet. A sketch of some of that structure (as we now imagine it) is shown below. We are relatively clear as to which nodes and links will exist, and somewhat clear on links' lengths (represented by geometric lengths in the diagram); however, a less settled issue is how to label the links. The sliplinks' lengths and labels in the sketch are intuitive rather than definitive. Each label simply says, more or less, what an average person would respond if asked, casually, what the two items it joins have in common. Certainly it would be more formal in the actual program. We anticipate that in Tabletop's Slipnet, sliplinks may well wind up having more than one label, so that they will shrink or expand under the simultaneous influence of a number of independent factors.

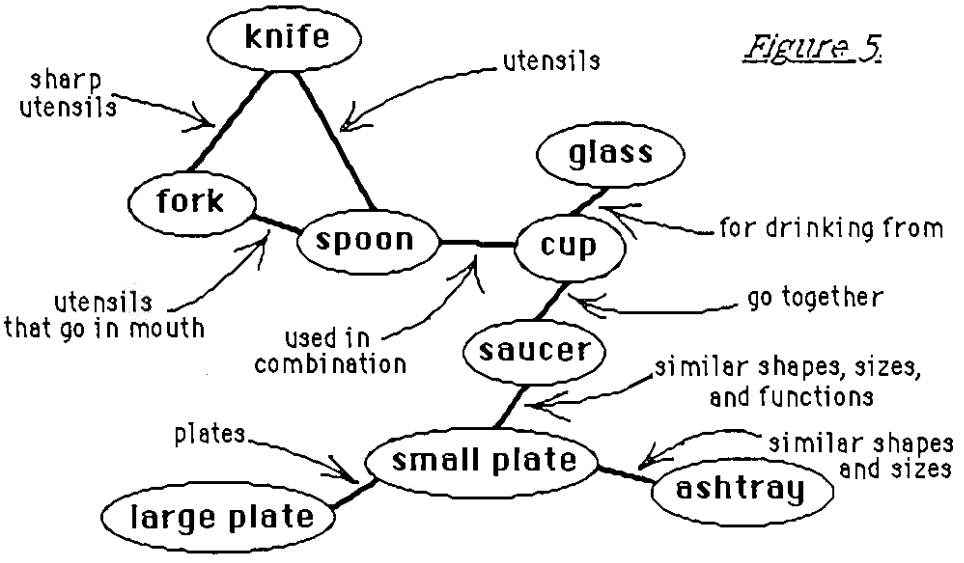


Figure 5

Interplay of Pressures: Two Examples

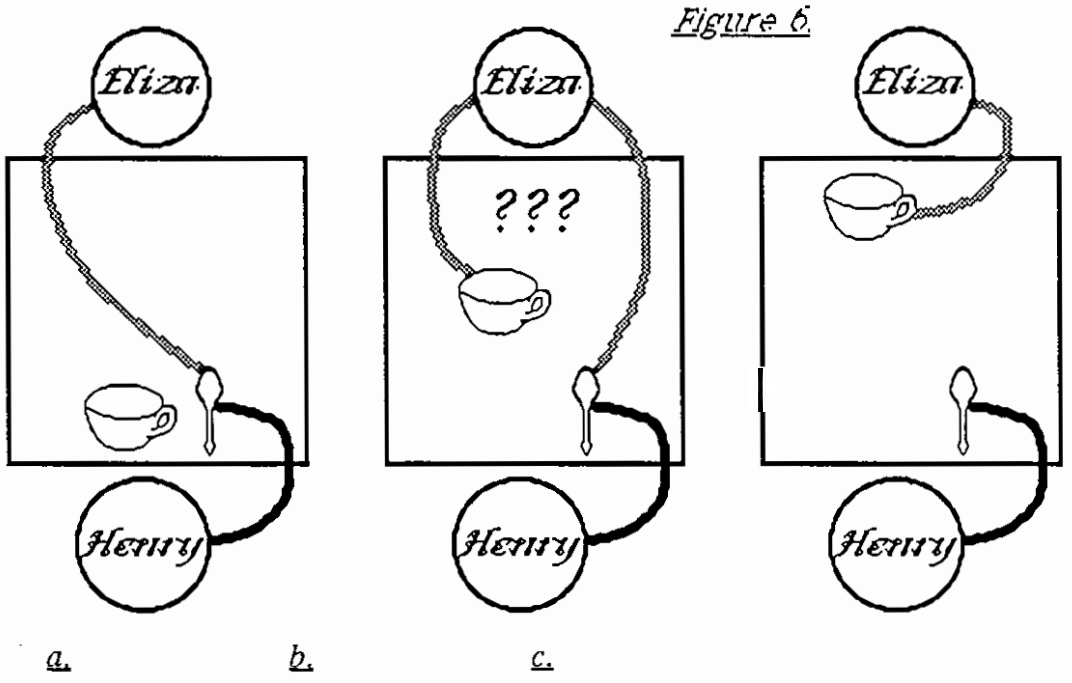
We have now briefly discussed pressures due to possession, conceptual proximity, and internal properties. The final notion, arguably the most subtle of the four, is that of physical proximity. To begin with, each individual has a somewhat ill-defined territory. Normally, one's territory corresponds to one's side of the table. However, personal territory can shift, depending on where one's personal objects are located. Roughly speaking, one's territory is where most of

one's objects are. If, for some reason, all of Eliza's objects are close to Henry, then her territory is Henry's side of the table. If someone has no objects, then their territory is their side of the table. Finally, both individuals' objects could be more or less uniformly spread over the table, in which case the notion of territory breaks down.

One of the major focal points of our research is, of course, to study how conflicting pressures give rise to different answers. Physical and conceptual proximity can interact and give rise to many fascinating and intricate pressure conflicts. Here is a very simple example involving territory and conceptual distance, whose purpose is merely to hint at the enormous potential complexity of the analogy puzzles that one can create in this domain.

Suppose that Eliza has no objects at all, and that Henry has a cup and a spoon directly in front of him (Figure 6a). Henry touches his spoon. In this case, Eliza will clearly touch Henry's spoon. But now imagine a continuum of situations in which Henry's cup is closer and closer to Eliza (suggested by Figures 6b and 6c); in each, Henry touches his spoon and says "Do this!"

Figure 6.



The perceived roles of the objects will gradually shift as the effect of physical proximity (specifically, the entry of the cup into Eliza's territory) becomes progressively more important, finally overcoming the opposing pressure due to the large conceptual distance between the objects. At that point, Eliza will touch Henry's cup, since its position, not its category, has become its dominant feature.

We should emphasize that Eliza responds to each new puzzle completely freshly, starting out with all the parameters in her Slipnet (i.e., node activations and sliplink lengths) at their default values. Thus, closely-related analogies forming a "family" such as was presented above are not intended to be a temporal progression of puzzles. However, we are also interested in temporal progressions of analogies, and we will be able to simulate such progressions by simply letting the Slipnet's final state after one puzzle be used as the initial state for another puzzle. This way, residual concept-activations and altered conceptual distances will bias the perception of the new situation and presumably give rise to psychologically plausible alternative answers. (This could be done also in Copycat.) Experimentation along these lines will provide a strong test of the generality

and psychological reality of our model.

Inter-object separation and distance from various edges of the table are other aspects of physical proximity that can become important in certain contexts. For instance, if Henry's knife and spoon are physically very close to each other (see Figure 7), they may be chunked into a perceptual unit by Eliza, and as such, become a landmark with respect to which she identifies the touched object -- say, Henry's cup, which she perceives as "the object closest to Henry's utensils landmark". If there is a similar "utensils landmark" in Eliza's territory, then the fact that her glass satisfies the translated description "the object closest to my utensils landmark" may make her wish to pick up her glass, even in preference to her own cup, which, though closely resembling Henry's cup, is not near the landmark. Obviously, the context in which this decision is made (other objects on the table, their locations and owners, etc.) will deeply influence Eliza's decision, and it is not by any means a foregone conclusion that in this situation, or in all close variants, she will choose her glass over her cup. However, there is a genuine pressure to do so, even if it does not win out, and the mere presence of that pressure may have "backstage" effects that help determine the eventual winner, much as a minor third-party candidate in a presidential election can, by siphoning off votes from the two major candidates, tilt the outcome in unpredictable ways.

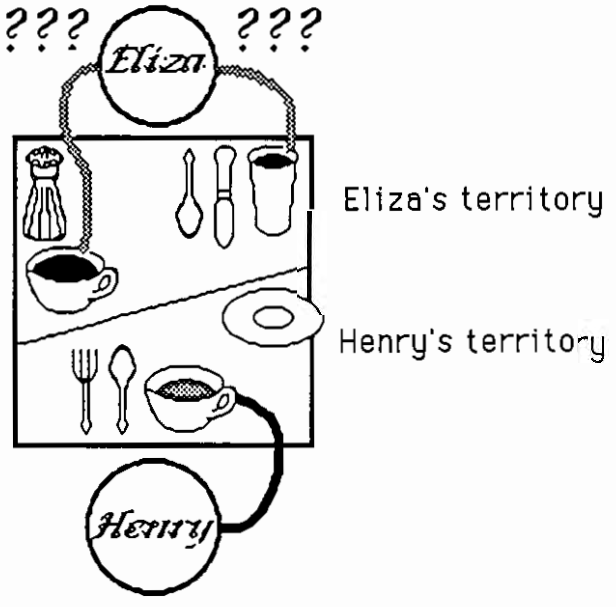


Figure 7

Novel Features of Tabletop

Tabletop's microworld is more chaotic than Copycat's alphabetic microworld, in the sense that it contains no orderly set of Platonic categories corresponding to the Platonic alphabet. Also, the one-dimensionality of Copycat's strings, although in principle as rich and potentially chaotic as any multidimensional space (one can code any N-dimensional structure into a long linear sequence), tends in fact to suggest problems featuring largely regular and symmetric structures, often even highly elegant ones. This is both an advantage and a disadvantage of Copycat's world. It is advantageous in that it encourages us to bring esthetic-judgment mechanisms into the architecture of Copycat and to let them play a large role, which brings us closer to both art and science. It is disadvantageous in that so much of the real world really is irregular and somewhat chaotic, and so Copycat problems often seem too artificial and pristine. In any case, the Tabletop world provides a distinct contrast to that, and when more than just a couple of objects are on the table, a very large number of forces enter into play. We find this richness of Tabletop to be an exciting challenge, and are convinced that our research in the Tabletop world will considerably expand our understanding of analogy-making mechanisms and fluid concepts.

Letter Spirit

The Key Role of Analogy in Determining Style

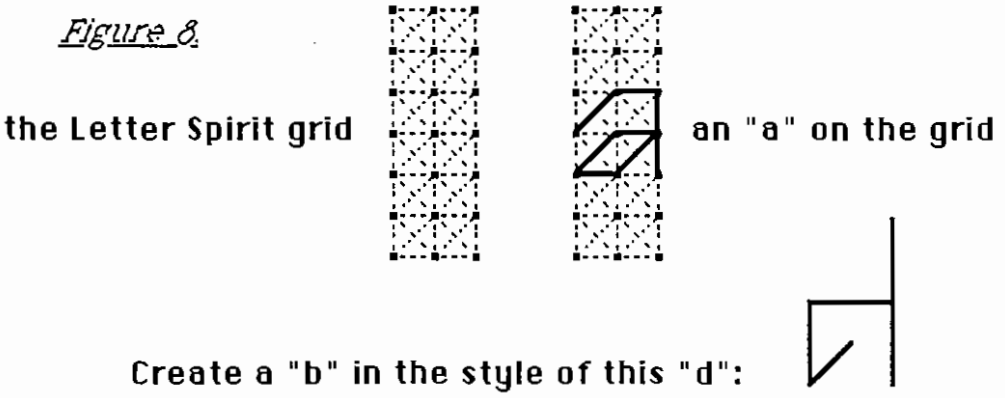
This project involves letters of the alphabet, but in a completely different way from Copycat. There, letters are abstract entities without shape or sound; here, by contrast, shape is everything. The Letter Spirit project has as its aim the modeling of how the 26 lowercase letters of the alphabet can be realized in a common style, so that they share a uniform "spirit". The idea is for the Letter Spirit program to start with a "seed" letter, and, using that letter as an inspiration, to create all the rest, one by one. Any letters that have been generated can exert stylistic influences on new letters.

Although the project focuses on style in one particular domain, we are interested in the general question of style. Thus we are driven by such questions as: What do all pieces of music by Bach have in common? How can one extend a style in lettering to a style in furniture design or architecture, or even music? What characterizes so broad a style as Art Deco?

We believe that the answers to all these questions have to do with abstraction and translation -- the key issues of analogy-making. The Letter Spirit project represents an attempt to render such profound questions approachable in a scientific manner. Obviously, we have had to simplify certain matters enormously, but the crux remains: propagation of an artistically uniform style through a broad range of distinct and highly variegated categories (i.e., the 26 lowercase letters).

To make matters as abstract and as conceptual as possible, we have bypassed all lower-level-vision aspects of letterforms by restricting shapes to the simple, discrete grid shown in Figure 8. (In this context, "lower-level vision" means the conversion of an array of dots into a representation involving line segments.) Letters must be formed as combinations of short line segments joining the dark points. An eight-segment "a" is shown on the grid to illustrate this idea.

Figure 8.

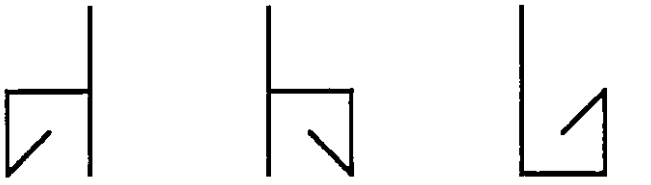


A typical analogy problem that arises from stylistic considerations is shown in the figure as well. The problem could be stated in this way: "Do to the category 'b' what has been done to the category 'd'." It sounds like a Tabletop "Do this!" command, and, as in Tabletop, there are many interacting pressures that combine to yield a good answer.

Interacting Pressures and Paradigm Shifts in Letter Spirit

What leaps to nearly everyone's mind is the mirror reflection of the given "d". This strategy, while usually a good basis for exporting style from a "d" to a "b", poses a special difficulty here, because the mirror-image shape (see below) probably belongs more to the category "h" than to the category "b". At best, it is a dubious "b" -- so we have encountered a snag. As in Copycat and Tabletop problems, one needs a loose, fluid way of transporting essence from one framework to the other, characterized by well-chosen slippages. Figure 9 shows an elegant solution to the problem:

Figure 9.



"d" poor "b" (too "h"-ish) good "b"

Recognition of the trouble with the first shape proposed requires a clear understanding of the essence of letter categories "b" and "h", and finding a slippage that yields an elegant solution requires a clear understanding of the spirit of the original "d". This type of problem can be characterized as a fight between two forces: an "attractive" force that wants to pull the shape as close to the center of the intended letter category as possible, and a "repulsive" force that wants to push the shape away from the center and toward the fringes of the letter category, to make the style as conspicuous as possible.

Brief Comments on the Letter Spirit Architecture

The Letter Spirit project is more concerned with details of the inner structure of categories than the previous two projects are. Although low-level vision is bypassed, high-level vision still must be simulated, which means the assemblage of line segments into meaningful wholes on various levels. Intermediate levels include "roles" such as "post" (the vertical element in any "d", "h", etc.) and "bowl" (the rounded part of a "d", "b", etc.). The highest type of meaningful whole is of course a complete letter, recognized for what it is (i.e., matched with its Platonic category). Figuring out which Platonic category is represented by a particular set of lines is very subtle in the Letter Spirit domain, because the shapes can belong to their intended categories in such exotic ways. This is illustrated in Figure 10, which shows the initial seven letters of seven different "spirits".

To extend each row all the way to "z" is a very challenging problem even for humans. Part of the problem is that one constantly runs into unanticipated snags (like the "h"-snag above), forcing one to conceive of the familiar letter categories in radically new ways. Another problem is that several already-created letters may simultaneously influence a human (or a program) trying to come up with a new "h" or "m" or "z". To deal with so many influences at once is one of the central difficulties of the Letter Spirit project.

The Letter Spirit Slipnet will contain nodes for all 26 Platonic categories, of course, with sliplinks indicating what two given letters have in common, such as the fact that often, "b" and "d" are mirror images, or the fact that an "h" tends to be like an "n" with a post. It will also contain nodes for all the roles inside letters, such as "post", "bowl", "arm", "crossbar", and so on, indicating cross-connections between them as well. Certain lower-level perceptual constructs will also appear in the Slipnet, such as "corner", "tip", "three-way junction", "crossing-point", and so on, since they are clearly so close to the roles one level of abstraction above them. However, even lower-level perceptual constructs will not appear, since they are more procedural than declarative.

It is probably fair to say that the problems in Letter Spirit are harder than those in Copycat and Tabletop; moreover, they require extremely creative slippages, often based on nearly inarticulable esthetic grounds. We therefore consider the creation of alphabetic styles on this simple grid to be an ideal domain for the study of truly creative artistic leaps.

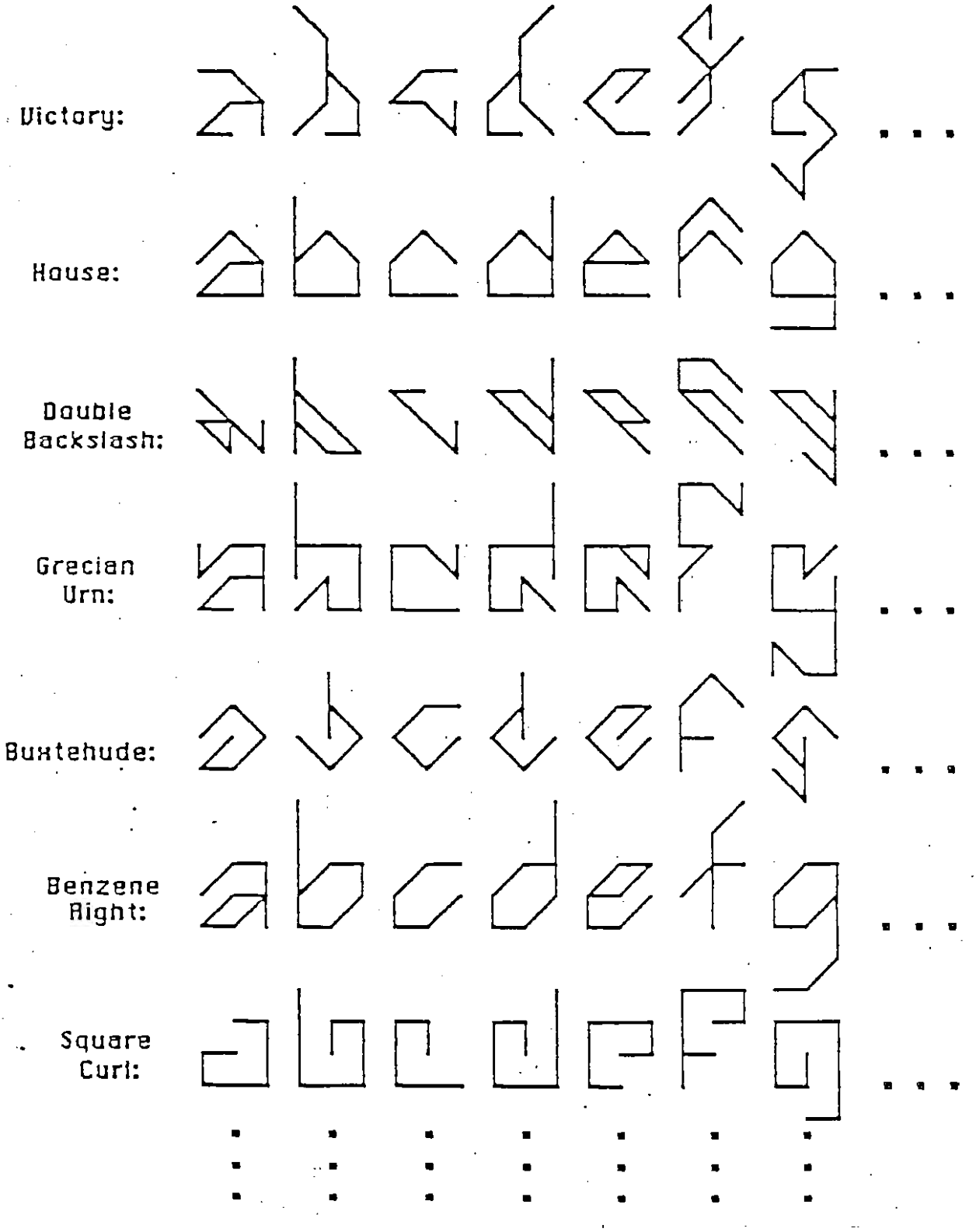

Victory:
House:
Double Backslash:
Grecian Urn:
Buxtehude:
Benzene Right:
Square Curl:

Figure 10.

Section III: Our Research in Perspective

Our Group and Its Work

Personnel of the Fluid Analogies Research Group

The "we" used in this section refers to the following people, constituting the Fluid Analogies Research Group at the University of Michigan and elsewhere:

Douglas Hofstadter (Professor of Psychology and Cognitive Science and Walgreen Professor of Human Understanding, University of Michigan);

Gray Clossman (computer-science graduate student, Indiana University);

Daniel Defays (Visiting Professor of Psychology, University of Michigan);

Robert French (computer-science graduate student, University of Michigan);

Greg Huber (computer-science graduate student, University of Michigan);

Alejandro López (psychology graduate student, University of Michigan);

Marsha Meredith (Assistant Professor of Computer Science, Blackburn College);

Melanie Mitchell (computer-science graduate student, University of Michigan);

David Rogers (post-doctoral research fellow in psychology, University of Michigan).

Hofstadter is directing the various projects, and the others (except Meredith) are working directly with him at the University of Michigan. Meredith has recently received her Ph.D. degree from the Computer Science Department at Indiana University.

Earlier Projects

We have already discussed the three major ongoing projects of our group: Copycat, Tabletop, and Letter Spirit. These three projects are closely related to two previous projects: Seek-Whence [Hofs82, Mere86] and Jumbo [Hofs83, Hofs86], which we now briefly describe.

Seek-Whence

Seek-Whence is a a discovery and extrapolation program; it tries to find the underlying regularity of a sequence of integers fed to it term by term -- in other words, to "seek whence" the sequence comes. Seek-Whence's numerical microworld is very similar to Copycat's alphabetic microworld, although there is a slightly different focus, in that the top-level task of the Seek-Whence program is to discover a rule rather than to create an analogy. However, we have found that the creation of analogies is just as indispensable to Seek-Whence as the discovery of regular patterns is to Copycat. Therefore, the two projects have many common goals and methods. The Seek-Whence program has been implemented by Meredith [Mere86]; it currently can solve fairly sophisticated sequences, using a probabilistic Slipnet architecture. The closeness of Seek-Whence and Copycat is explained in [Hofs85a], where a host of analogy problems that cropped up in Seek-Whence are presented; [Hofs82] traces the motivation and development of the Seek-Whence project in detail. (See also the comments under "Meredith" in the subsection "Comparisons with Other Work".)

Jumbo

The Jumbo project [Hofs83, Hofs86] was undertaken expressly as a prelude to the implementation of the more ambitious Seek-Whence and Copycat projects. Its purpose was twofold: (1) to study a specific type of parallelism inspired by the architecture of the Hearsay-II speech-recognition project [Redd76] as well as by the metabolism of the living cell, and (2) to model the ability of humans to fluidly group, take apart, and internally restructure components of

all types of ideas -- an ability that we feel is a key element of creativity. The domain of Jumbo is that of short anagram puzzles called "Jumbles". The program tries to create plausible word-candidates from a jumbled set of letters. It has no dictionary in which to look up words; instead, it evaluates its solutions purely on internal criteria of what makes a good word-candidate. Jumbo's knowledge of English consonant clusters and vowel groups is built-in.

The goal of this project was to model the mechanisms underlying the rapid assembly, disassembly, and regrouping of letters that go on unconsciously in the mind of a human solving Jumbles. It is our belief that similar "shuffling" mechanisms underlie the creative manipulation of ideas in many domains. The architecture of Jumbo was a precursor to that of our current projects. It is based on the notion of the parallel terraced scan, in which many pathways are explored simultaneously, at rates roughly proportional to their estimated promise. Fundamental to Jumbo are the operations of chunking (grouping), and of regrouping while maintaining "well-chunked wholes" (e.g., "pan-gloss" can be regrouped as "pang-loss" without being reduced to the letter level). These notions are extremely important to our current work as well. Jumbo also used a computational temperature, which, as in our current projects, is a measure of how close the system "feels" it is to a good solution. The higher the temperature, the more disorder is to be injected into the system, which allows the system to explore as-yet-untried pathways, thus rendering its search for a solution more efficient. Jumbo's category system was quite meager (it contained only a few categories, such as "cluster", "syllable", etc., with very simple relationships among them), and it had no Slipnet.

The Jumbo program served its purpose of testing out certain crucial architectural ideas. Once it had been completed (July 1982), Meredith was able to move ahead quickly with her implementation of Seek-Whence, and Hofstadter was able to design the basic architecture of Copycat.

Publications

Aside from the publications on Seek-Whence and Jumbo cited above, we have written a number of articles describing aspects of our research. On Copycat, there are the following:

"Simple and Not-So-Simple Analogies in the Copycat Domain" [Hofs84a];
"The Copycat Project: An Experiment in Nondeterminism and Creative Analogies" [Hofs84b];
"De basis van het 'Copycat'-project" (in Dutch) [Hofs84c];
"A Nondeterministic Model of Analogies, Based on the Ising Model of Ferromagnetism" [Hofs87].

On intensionality, slippability, roles, and analogies, there are these:

"Shakespeare's Plays Weren't Written by Him, But by Someone Else of the Same Name" [Hofs80];
"Analogies and Roles in Human and Machine Thinking" [Hofs85a].

On fluid concepts and creativity, there are the following:

"Variations on a Theme as the Crux of Creativity" [Hofs85c];
"On the Seeming Paradox of Mechanizing Creativity" [Hofs85d];
"Metafont, Metamathematics, and Metaphysics" [Hofs85e].

On statistically emergent models of mind, there are:

"Waking Up from the Boolean Dream, or, Subcognition as Computation" [Hofs85b];
"Who Shoves Whom Around Inside the Careenium?" [Hofs85f].

More recently, we have written and submitted the following articles to conferences:

"A Theory of the Creation of Fluid Analogies, and its Computer Implementation" (submitted to the 1987 International Joint Conference on Artificial Intelligence);
"Fluid Concepts and Creative Analogies: A Theory and its Computer Implementation" (submitted to the 1987 Cognitive Science Conference).

We have written up a progress report called "FARG Status Report, February 1987" that summarizes where our computer projects are now and where they are going, and relates them to several broader-based somewhat informal studies of human creativity that have been ongoing in our group for a number of years.

Finally, it is worth pointing out that in Chapter 19 of Gödel, Escher, Bach [Hofs79] , Hofstadter sketched out the architecture of a hypothetical system to solve Bongard problems [Bong70]. The system described there included the notion of a Slipnet and hinted at the advantages of a nondeterministic control structure, although that notion was not fully realized until the Jumbo project was implemented. We still feel that Bongard problems constitute one of the best domains ever proposed for AI research, and hope that our current work will help advance the possibility of such work.

Comparisons with Other Work -- A Detailed Discussion

In the following several pages, we compare our work in detail with that of many other research teams. Most of them are (or were at one time) working directly on analogy, but some have a different focus that we find relevant to our work. We have striven to include every relevant group, but undoubtedly have omitted some inadvertently. Groups whose primary focus is learning or induction were generally excluded, unless their methodology or goals seemed especially close to ours. The groups included (alphabetized by principal investigator's name) are:

John R. Anderson
Mark Burstein
Jaime Carbonell
Lindley Darden
Thomas Evans
Jerome Feldman
Dedre Gentner
Russell Greiner
John Holland
Keith Holyoak
D. Kahneman and D. Miller
Pentti Kanerva
Smadar Kedar-Cabelli
Patrick Langley et al
Douglas Lenat
Marsha Meredith
J. Moore and A. Newell
Donald Norman
Walter Reitman
D. Rumelhart, J. McClelland, et al
Roger Schank
Patrick Winston

Anderson

John R. Anderson believes, as we do, that spreading activation plays a central role in controlling cognitive processes. For this reason, several parallels can be drawn between the architecture of his ACT* [Ande83] and that of our systems. The declarative memory of ACT* is a network with spreading activation, and in that sense resembles our Slipnet, but there is one major

difference: the nodes in our Slipnet represent exclusively concepts, whereas those in ACT*'s declarative memory are often abstract propositions -- and occasionally even spatial images. ACT*'s production memory is comparable to our repertoire of tasks, although our tasks carry out a broader range of operations than the condition/action production rules that populate ACT*'s production memory. Anderson's productions are activated by pattern-matching tests that are performed repeatedly at varying rates determined by the activation of related nodes in declarative memory. This effectively accelerates certain types of productions in response to interacting pressures. This closely resembles our notion of the activations of Slipnet nodes affecting the urgencies of tasks, thereby translating pressures into differential rates of exploration of pathways. Whereas Anderson has a mechanism for transfer of short-term memory structures into long-term memory, we have no counterpart, because for the time being we are not concerned with learning.

Burstein

Mark Burstein's program CARL [Burs86] learns and makes inferences about assignment statements in BASIC by using the analogy "variables are like boxes". Although Burstein's focus is somewhat different from ours (he is modeling results of psychological studies of how people use analogies to reason about new domains), many of his concerns parallel our own. He is interested in how context determines whether mappings between the source and target domains are based on similarity between extrinsic roles or between internal properties of objects, and how context determines which roles or features are used in the mappings. His preference for abstraction is reflected in his system's use of heuristics similar to Gentner's "systematicity" conditions and in his use of context-dependent "levels of abstraction". Burstein's notion of mapping base-domain relations to "sibling" target-domain relations is similar to, but less central than, our notion of "slippability" between concepts close in the Slipnet.

Carbonell

Jaime Carbonell's work on analogy [Carb83] is concerned with how a new problem can be solved by retrieving and transforming the solution to a similar, previously-solved problem. His approach is quite different from ours: he models the analogical transformation as problem-solving by means-end analysis in a state space. We share with Carbonell the basic view that analogy is central to cognition, and his work reflects commonality with ours in its concern with questions of similarity, generalization, and the abstraction of essential aspects of situations. His earlier research on metaphor [Carb80], using what he calls a "hierarchy of variability" (we might call it "levels of slippability"), parallels our concern with what slips most easily versus what is deepest in a given structure or situation.

Darden

Lindley Darden has been working on understanding analogies from the point of view of a philosopher of science [Dard83]. She concentrates on very large-scale real-world analogies that have inspired the development of major scientific theories. We are very interested in this kind of approach, and hope that her work and ours will converge on similar basic principles.

Evans

Thomas Evans explicitly recognized the importance of analogy-making in his attempt in the mid-1960's to implement a system that could handle geometric analogies. His program's object-to-object mappings and primitive rule-translations foreshadow similar operations in Copycat. However, Evans had only an incipient notion of concept-to-concept mappings, and nothing resembling a Slipnet embodying conceptual distances. Moreover, his program's control structure was strictly deterministic, proceeding strictly through predetermined stages. Despite these fundamental differences, we share with Evans the strong belief that analogy can be studied very

well in completely abstract domains in which there are no purposes (e.g., for problem-solving) served by the analogies that are constructed. It seems that humans simply share an intuition about what makes for a deep mapping, and what makes for a shallow one. The abstract microworld of geometric analogies used by Evans is closely related to the extraordinarily rich world of Bongard problems [Bong70, Hofs79], one of the early inspirations for the Copycat architecture.

Feldman

The connectionist work by Jerome Feldman and colleagues [Feld82] concentrates mostly on low-level visual problems. Their "stable coalitions" of small computational elements are similar to regions in a Slipnet that reach equilibrium activation-levels under the combined influence of mutual reinforcement, mutual inhibition, and decay. Feldman concentrates on levels of cognition considerably below those on which our research focuses. Nonetheless, we believe that many of the lower-level mechanisms discussed by Feldman adapt quite well to the more cognitive level at which we are working. In particular, we rely heavily on spreading activation (what Feldman calls "mutual reinforcement") and decay ("habituation").

Gentner

Dedre Gentner has investigated the question "What makes an analogy powerful to humans?" Her theory [Gent83] proposes that what makes an analogy strong is preservation of higher-order relations ("systematicity") with less importance paid to lower-order relations, and even less to unary properties possessed by objects in the two domains. While we fully recognize the appeal of such a clean answer to the enigma of analogy, we feel it is only one aspect of a much larger truth. Our programs take into account many battling pressures in coming up with an answer, of which Gentner's drive toward systematicity is an important one. That pressure is represented largely in our work by the high degree of "semanticity" assigned to certain very abstract concepts ("symmetric-opposite", for instance), which tends to push hard for analogies based on higher-order relations. In other words, in our systems, a drive toward systematicity is emergent, as distinguished from being explicitly incorporated as a driving principle. A second way in which systematicity emerges from many pressures in our work is the necessity of bridges to be compatible, which means that all slippages must be close to parallel. The uniform "direction" of several parallel slippages amounts to a single very abstract constraint guiding lower-level aspects of the analogy, and such a constraint is precisely what a higher-order relation is. On the other hand, the fact that in a very broad sense we agree with the idea of systematicity in analogy-making does not imply that all analogies produced by our systems will favor systematicity over all else. In many cases, the pressures pushing for systematicity are simply overwhelmed by other types of pressure, producing analogies that have commonsense appeal but that nonetheless are anomalous from the point of view of Gentner's theory.

Greiner

Russell Greiner's Ph.D. thesis [Grei85] presents a logic-based, heuristic-search approach to analogy-making. His program is given a set of predicates and relations about objects in the source and target domains, and is told initially that *object1* in the source domain is "like" *object2* in the target domain. Its task is to find an "abstraction" (a pre-formed abstract set of relations, portable from one domain to another) that can be instantiated by objects in the source and target domains. Greiner's methodology is quite different from ours: his program uses heuristic search to find an abstraction and instantiations for the abstraction, and logic to prove that the instantiations are consistent with the program's permanent information. His goals are also somewhat different: he is interested in how the notion of a "useful" analogy (that is, useful for solving a given problem) can be formalized and implemented in heuristics. Greiner's general conclusions about analogy-making have something in common with our view: his program, like ours, prefers abstract ways of looking at situations and simplicity in describing situations. His system uses heuristics that

sometimes contradict each other, similar in some sense to our competing chains of tasks, reflecting the shared view that various pressures compete to make an analogy.

Holland

The work on classifier systems by John Holland and his colleagues [Holl86] is surprisingly close in spirit to our work. Holland's careful study of the "two-armed bandit" problem [Holl75] led him to many of the same conclusions as we have reached about the importance of a stochastic element in modeling intelligence. Like us, Holland stresses the statistical emergence of concepts from a boiling "soup" of subcognitive actions. His systems, as one would expect from their name, place great weight on classifying the environment, and this is achieved by activation of a coherent set of "classifiers" (production rules expressed as bit strings). Strength of a classifier maps roughly onto our activation of a concept node, and the coexistence of multiple pressures, achieved in our systems via the simultaneous activation of many nodes and interleaved chains of tasks, is achieved in his systems via the parallel firing of many rules. A category is distributed over numerous classifiers, as for us it is distributed over numerous Slipnet nodes. In fact, the overlapping nature of categories, a fairly explicit property of our Slipnet, is somewhat more implicit (i.e., emergent) in classifier systems, and by the same token, spreading activation, which we directly program, is also an emergent property of classifier systems. Thus, several properties that we implant directly into our systems emerge automatically in classifier systems, which is an admirable achievement; however, the fact that the basic operations of classifier systems are much further from the cognitive level makes the problem of controlling their high-level behavior much more difficult.

Holyoak

From their psychological studies of analogy, Keith Holyoak and his colleagues have formulated a theory [Holl86, Gick83] in which we see much that resembles our own work. In their view, as in ours, analogies are created by the interaction of many competing top-down and bottom-up pressures, influenced by the context of the analogy and various types of resemblances between the source and target situations. They propose that an initial "partial mapping" is made between source and target, and that this partial mapping influences the subsequent extended mapping in a top-down fashion. This is very similar to our view of how a coarse-grained worlds-mapping mediates rule translation. Their model is different from ours in that they propose that, from the partial mapping, an abstract "schema" is formed in which each pair of corresponding objects and actions is assigned an abstract category describing both analogues. In our system, no such explicit overarching static schema is formed; instead we produce a dynamic structure (the rule) that fluidly changes as it migrates from one situation to another. This has even more generality than a schema, we feel. We see this as an interesting difference between the two approaches that deserves future attention. Holyoak and colleagues are also concerned with how situations analogous to a given situation are retrieved from memory. Their model suggests that a given situation has an "aura of associations" that triggers, through spreading activation, memories of similar, possibly analogous situations. This is quite similar to our view of the associative nature of concepts, as embodied in the Slipnet. Finally, we find that Holyoak's experimental methodology is similar to ours: just as we "tune" the analogy problems we give to our system to see how the strengths of various pressures affect the program's behavior, Holyoak and colleagues did experiments in which human subjects were presented with variants of a certain analogy, each designed to bring out different pressures, in order to study such effects.

Kahneman and Miller

Daniel Kahneman and Dale Miller's recent work on norm theory [Kahn86] has numerous points of contact with our work. "Normal" features of an event, according to them, correspond to one's expectations or prototypes, whereas "abnormal" features deviate radically from one's expectations. An abnormal feature of an event, they claim, is likely to "slip" to its normal counterpart in a prototype evoked by the event. Their emphasis on the importance of understanding the mechanisms underlying slippages -- a process we feel is at the heart of human thought -- makes their work close in spirit to ours. Of particular interest to us is their theory of counterfactuals and mechanisms of counterfactualization. People reveal "hingepoints of thought" when they spontaneously imagine counterfactual variants of situations they are facing. Kahneman and Miller's study of the types of counterfactuals people actually manufacture (as distinguished from "counterfactual counterfactuals" -- ones that people might manufacture but do not) represents a significant step toward understanding conceptual slippage from the viewpoint of cognitive psychology. We hope that their psychological studies and our modeling work will enrich each other.

Kanerva

Pentti Kanerva has developed a theory of robust memory (i.e., memory in which slight degradation of an address does not degrade retrieval), based on a distance measure between pairs of long bit-strings [Kane84]. Although his work is not directly related to ours, its stress on statistics for retrieving the "essence" of a situation is very consonant with our views. Kanerva's work suggests a lower-level manner of implementing a Slipnet, in which conceptual distance is so important.

Kedar-Cabelli

Smadar Kedar-Cabelli has developed a system to carry out "purpose-directed analogy". The idea is that since any two situations have many ways of being mapped onto each other, one needs to know the purpose for which the analogy is being made to select the appropriate mapping. We certainly agree that there are many possible mappings, and that selection is necessary. We also agree that sometimes, knowledge of purpose helps. However, we disagree on the centrality of purpose. Many of our analogy problems have compelling answers despite the absence of any purpose they might serve. This is because humans have a strong sense of context-independent similarity; that is, certain mappings seem compelling in and of themselves (consider the geometric analogies in [Evans68], for instance). We are interested in seeing how differences in pressures can shift the preference of one mapping over another. By adjusting features internal to a given analogy, we can vary pressures, and thus make certain mappings more or less desirable. Our pressures do not include purpose, though, because we are not working on analogy in a context of problem-solving. In many of our analogies, considerations of esthetics and simplicity, rather than purpose, play a central role. Our approaches differ along several other dimensions as well. Kedar-Cabelli's approach is logic-based; membership in a category is established by resolution theorem-proving methods. To us, logic seems too rigid a basis to provide the underpinnings of an approach to fluid analogy-making. Another discrepancy between our approaches is that we see no analogue in Kedar-Cabelli's system of our Slipnet, with its conceptual distances and potential slippages.

Langley et al

Patrick Langley and several co-workers, including Herbert Simon, Jan Zytków, and Gary Bradshaw, have for several years been working on induction and scientific-theory formation [Lang83]. Their earliest project, Bacon, has had several successors, including Glauber, Dalton, and Stahl. Bacon represented a rather bottom-up approach to scientific discovery, being primarily concerned with finding a mathematical formula to encompass all the data in a given set, without

regard for how that formula fit into any overarching theory. The later projects have been more concerned with the manipulation of concepts. Potentially, this work has much in common with ours, but up till now, analogy-making has not been seriously treated in their work.

Lenat

Douglas Lenat's AM and Eurisko projects [Lena83a, Lena83b] are motivated by many of the same questions as our projects are. We share an interest in, and many intuitions about, such topics as discovery in mathematics in science, relations between intelligence and evolution, the abstract nature of heuristics, and epistemological questions. The fundamental difference is that Lenat believes that a purely high-level approach (i.e., heuristics, meta-heuristics, etc.) is the royal road to discovery and creativity, while we believe somewhat the opposite -- namely, that the central problem is the nature of concepts, and therefore that the heart of the matter is below the conceptual or cognitive level, on the subcognitive level. Thus we tend to stress emergent behavior produced by competing low-level pressures, while Lenat looks for high-level principles that resolve conflicts, and even higher-level principles that resolve conflicts between high-level principles. Consequently, Lenat's work is not concerned at all with parallelism, statistics, or emergent behavior, in distinct contrast to ours.

Meredith

Marsha Meredith recently completed her Ph.D. thesis under the direction of Douglas Hofstadter [Mere86]. It describes her program, called "Seek-Whence", which accepts terms of an integer sequence one at a time, attempts to formulate a rule accounting for what it has seen, and either verifies or refutes its hypothesis as it sees the next actual term. The sequences the program is aimed at are not mathematical but highly patterned (e.g., "1-2-2-3-4-4-5-6-6-...." and "2-1-2-2-2-3-2-4-2-5-...."), and often resemble melodies with simple or complex rhythms. If the hypothesis for a sequence is refuted, the system attempts to modify it by using a probabilistic Slipnet-type architecture very closely related to those of Copycat and its predecessor, Jumbo [Hofs83, Hofs84b]. Meredith's work is distinctly psychological, as opposed to the majority of earlier sequence-extrapolation work [Piva64, Pers66]. Rather than exploiting a host of specialized mathematical techniques (e.g., taking differences and ratios, looking at subsequences formed by taking every n th element with n =2, 3, 4,..., and applying the program recursively to derived sequences), Meredith uses an approach based on perception, in which local links are formed, suggesting local "gloms" (groups), which form and burst in a kind of "soup" of activity, until stable units gradually emerge and reinforce each other, suggesting a conceptual organization behind the scenes.

Moore and Newell

The "Merlin" project [Moor74] was undertaken by James Moore and Allen Newell in the early 1970's. Its premise that a kind of conceptual mapping underlies all thought is entirely consonant with our view of analogy as the driving force behind cognition. Moore and Newell attempted to make a "recursive" analogy-maker, in which a high-level mapping would "force" lower-level mappings until the process bottomed out in some already-known mappings. It was a fascinating idea but was never fully implemented, and seems not to have been followed up.

Norman

Donald Norman's work on explaining human errors [Norm81] and his joint work with David Rumelhart on simulation of an error-making human typist [Norm82] are both based on a parallel architecture in which "schemas" with varying levels of activation compete for control of behavior. The ideas in these projects had a good deal of influence on the development of the Slipnet architecture.

Reitman

Often overlooked, the work of Walter Reitman on analogy [Reit65] -- specifically, his "Argus" project -- is a forerunner of many currently popular ideas. Reitman was one of the first to see how simulated parallelism could allow competing pressures to coexist, in the form of simultaneously activated concepts that controlled the flow of a program. His program was a reaction against logic-based strategies for cognition, like Simon and Newell's General Problem Solver, and was an attempt to come much closer to true mechanisms of human cognition. Although very limited, Argus pioneered ideas that are only now being rediscovered.

Rumelhart, McClelland, and the PDP research group

Many of the key ideas of David Rumelhart and James McClelland [McCl86] and colleagues agree with our own, giving our work a distinctly, although not uniquely, PDP ("parallel distributed processing") or "connectionist" flavor. Their work, like ours, is based on a bottom-up, statistically emergent approach to cognition. They, like us, stress probabilistically-sequenced subcognitive events that coalesce into larger "molar" events -- the stuff of conscious perception. Particularly germane to our research are some of the following ideas and their proponents: (1) Statistical emergence: This is the fundamental faith of connectionism -- that cognitive phenomena emerge as statistical effects of a large number of small, distributed subcognitive events with no global executive. In contrast to much of the PDP work, however, we are interested in applying many of the principles of connectionism to networks of nodes at a nearly-cognitive level (Slipnets). (2) Parallelism: Two central notions motivated the parallelism in our architecture: the inherently parallel nature of perception (which we feel is close to the core of cognition), and the pervasiveness of cognitive phenomena explainable only through some type of spreading activation. While we do not believe "massive parallelism" to be a panacea, we do believe that perceptual processes and spreading activation are inherently parallel, and are keys to the fluidity of human cognition. (3) Nondeterminism and temperature: The idea of a probabilistic substrate to cognition is a strongly shared intuition. In Geoffrey Hinton and Terrence Sejnowski's Boltzmann machines [Hint83], a representation of the environment is produced by probabilistically modifying the weights of connections between the nodes of a net as the environment is scanned iteratively. Some of our small probabilistically-scheduled tasks are similar to their feature detectors, and their modification of inter-node weights is reminiscent of our modification of inter-node distances in the Slipnet. Their search for near-optimal states relies, as does ours, on a notion of computational temperature, developed independently by several different researchers [Hofs83, Kirk83, Hofs84b, Smol84]. In our architecture, increasing the temperature is the main way for breaking structures that have proven of little use. The notion of "happiness" developed in Hofstadter's Jumbo program is similar to Paul Smolensky's notion of "harmony" [Smol84]. (4) Competition: One key feature of our architecture -- the way bridges engage in a winner-take-all struggle in the worlds-mapping phase -- is reminiscent of Rumelhart and Zipser's unsupervised-learning paradigm, which they call "competitive learning". A point of difference with their model is that lateral inhibition is, for the time being, absent from our work.

Schank

Roger Schank and his students and colleagues at Yale have numerous projects that overlap with ours in interesting ways. One of Schank's most recent interests is the roots of creativity, which, in common with us, he seems to feel have much to do with analogy. He is interested in what he calls "reminding incidents", and what we might call "deep analogical retrieval" (i.e., the retrieval of memories through highly abstract cues), and the mapping of retrieved memories onto current situations in fluid ways, allowing understanding of the current situation by means of a kind of "translation" or "modification" of the past memory. His student Michael Dyer, in his Ph.D. thesis [Dyer84], explored the extraction of such "deep essences" from short narratives; there, the essences were called "adages" and "thematic abstraction units". A major methodological

difference, however, is that parallelism and statistical emergence have no place in Schank's approach. Nor do we find anything resembling a Slipnet (conceptual overlap and conceptual distance) in the complexities of Schank's memory models based on MOPs (memory organization packets) and TOPs (thematic organization packets). We feel that over the years, there has been an increasingly large overlap of interests between our work and Schank's, and we hope for some opportunities for communication with Schank and colleagues.

Winston

Patrick Winston [Wins83] is concerned with importance-weighted mappings of roles in one situation onto roles in another. His approach to determining "importance" is based on counting the number of constraints in which the given entity participates. This would seem, roughly, to weight abstract roles more heavily than superficial ones, a tendency that is consistent in outcome, if not in method, with our approach. However, as he points out in the cited paper, in his work there are no clear principles about how pressures representing different levels of abstraction interact with each other. Since this is one of our key foci (shared with Gentner), we find Winston's work conceptually quite distant from ours, especially insofar as his implementation employs a strictly top-down deterministic control structure, thus eschewing the idea of statistically emergent concepts.

Conclusion: Our Research in Perspective

First, a few words on what we are not trying to do. An important aspect of analogy that our projects do not deal with is how people retrieve memories in order to construct an analogy with a given situation [Carb83, Gick83, Scha83] ; in Copycat and Tabletop, at least, the two situations are pre-specified. (In Letter Spirit, matters are more complex, since, when trying to create a "z" given all 25 other letters, one can be influenced by any number of those already-created letters.) We are also not trying to make our programs create new concepts (i.e., to add new nodes to the Slipnet). In that sense, we are not modeling learning. On the other hand, our work does involve learning, if that term is taken to include the automatic generalization from experience that humans perform in novel contexts [Holl75, Holl86].

In all three of our projects, perhaps most obviously in Letter Spirit, esthetics plays a major role. We feel that the role of esthetic pressures in ordinary thought has been somewhat neglected. Not only in science and art, but even in daily life, people are unconsciously affected by considerations of simplicity, balance, symmetry, pattern, and so on. Such matters exert significant and pervasive, though often unconscious, pressures on judgments. We hope that our models will shed some light on how this comes about, and on the nature of esthetics.

We are attempting to create mechanisms for analogy-making in its full generality and in any domain. We are learning how to build a Slipnet whose concepts act with fluidity: stretching under pressure, adapting to novel contexts, triggering relevant associations, etc. We believe the mechanisms we are developing are psychologically realistic; the ultimate test is, of course, how well they mirror human performance over a broad range of problems. We feel the most important task for artificial intelligence and cognitive science is to explicate, via plausible working models, the nature and interaction of human concepts, and that is our focus.

Appendix: A Sampler of Copycat Analogies

The following is a collection of analogy problems in the Copycat alphabet domain. The idea is that in solving these problems, you should use only knowledge that Copycat has. Thus, assume you have no knowledge of letters' shapes or sounds, of words, or of any other linguistic or graphic facts. You know the alphabet equally well forwards and backwards, but only locally: for any letter, say **Q**, you know its immediate predecessor and successor (**P** and **R**), but you do not know: its ordinal position (e.g., **Q** is the 17th letter), which half of the alphabet it is in, or even how far it is from other letters (e.g., from **Q** to **S** is 2 steps, from **Q** to **M** is -4 steps). You do know that **A** is the initial letter and **Z** is the final letter; being extremities, they are landmarks of equal importance. The alphabet is not circular: **A** has no predecessor and **Z** has no successor. You also have a commonsense notion of grouping by sameness or by alphabetical order (forwards and backwards), and can recognize symmetry on a concrete as well as an abstract plane. You cannot do arithmetic, and even resist counting much higher than 3 or so. In general, you know that it is important to describe things economically, simply, elegantly, with a minimum of arbitrariness. You know that abstractions are good, that being overly literal-minded is bad. You also know that unjustified generalization is bad.

The problems given below are arranged into ten "families", each having a common idea or theme among its problems. All the problems are of the form "If S1 changes to S1', how does S2 change?", and are written out symbolically this way:

S1 ---> S1' (example)
S2 ---> ? (target)

In many of the families, several problems in a row are based on a single example. In those cases, the example is given only once, and the various targets are listed below it.

1. **abc ---> abd**
   - *a.* **ijk ---> ?**
   - *b.* **ijc ---> ?**
   - *c.* **ixjxkx ---> ?**
   - *d.* **ijkijklijklm ---> ?**
   - *e.* **kji ---> ?**
   - *f.* **cba ---> ?**
   - *g.* **cab ---> ?**
   - *h.* **pct ---> ?**
   - *i.* **aaabbbccc ---> ?**
   - *j.* **aaabbbcck ---> ?**
   - *k.* **ace ---> ?**
   - *l.* **xyz ---> ?**
   - *m.* **abcabc ---> ?**
   - *n.* **abcijk ---> ?**
   - *o.* **abcdef ---> ?**

2. *a.* abc ---> abe
   ijk ---> ?

   b. abc ---> abf
   ijk ---> ?

   c. abc ---> abq
   ijk ---> ?

---

3. abc ---> ccc
   a. ijk ---> ?
   b. xyz ---> ?

---

4. abcd ---> abcde
   a. ijklm ---> ?
   b. ijxlm ---> ?
   c. ijkpqr ---> ?
   d. mlkji ---> ?
   e. iiii ---> ?
   f. iiiijjjj ---> ?

---

5. *a.* mmmkooeeeeefqxx ---> kfq
   riipppppppplooyg ---> ?

   b. rrccmmkppbb ---> k
   ljooooosrezv ---> ?

---

6. *a.* abcde ---> xxxxx
   pqr ---> ?

   b. pqrxxxx ---> pqrstuv
   efghxx ---> ?

   c. axcxexg ---> abcdefg
   wxyx ---> ?

d. xxh ---> fgh
pxxx ---> ?

e. x ---> c
1. xy ---> ?
2. xyz ---> ?

---

7. *a.* eeeeqee ---> eeeeree
sosss ---> ?

b. eeeqee ---> qqqeqq
sosss ---> ?

c. eqe ---> qeq
1. abcdcba ---> ?
2. aaabccc ---> ?

---

8. *a.* abc ---> qbc
1, ijk ---> ?
2. xyz ---> ?

b. abc ---> zbc
xyz ---> ?

---

9. *a.* abcdde ---> abcde
pqstu ---> ?

b. abced ---> abcde
ppqqrrs ---> ?

---

10. *a.* a ---> z
b ---> ?

b. pqr ---> z
rqp ---> ?

c. **pqr ---> rqp**
a ---> ?

Plausible Answers

There are several plausible answers to many of these problems. Rather than listing them all, we have given what we believe to be the strongest answers, and occasionally we have given two or three if we consider it a very close call. There is, of course, room for disagreement and debate in certain cases.

1. *a.* **ijl**; *b.* **ijd**; *c.* **ixjxlx**; *d.* **ijk-ijkl-ijklmn** (chunked for ease of reading; here **ijklmn** is seen as the successor of **ijklm**); *e.* **lji** or **kjh**; *f.* **dba**; *g.* **dab**; *h.* **pdt** or **pcu**; *i.* **aaabbbddd**; *j.* **aaabbbddd** or **aaabbbddk** or **aaabbbccl** or ??? (here the question is what to do about that pesky **K**); *k.* **acg** (not **acf**, which doesn't take into account the "double-successor" structure of the target string, in which **C** is seen as "the successor of the successor of **A**", etc.); *l.* **wyz** (see the discussion of this problem in Section II, pages 11-12); *m.* **abdabd**; *n.* **abdijl** (the pressure to change both successor groups seems stronger than the pressure to change just one of them); *o.* **abcdeg** (it would seem artificial to perceive the target as **abc-def**, and then to treat it as the preceding one was treated, giving **abddeg**).

2. *a.* **ijm** or **ije** (should the **E** be seen as just "an **E**", or as "the successor of the successor of **C**"? Both views seem about equally plausible); *b.* **ijf** (**ijn**, based on seeing the **F** as "the successor of the successor of the successor of **C**", seems rather implausible); *c.* **ijq** (here there is no question: there is no relationship between **C** and **Q**).

3. *a.* **kkk**; *b.* **zzz** or **xxx** (since **A** and **Z** are "distinguished" letters, **abc** and **xyz** resemble each other in reverse order, with the **A** corresponding to the **Z**, and the **C** to the **X**).

4. *a.* **ijklmn**; *b.* **ijxlmn** (extend the "successor group", even though it is imperfect) or **ijkxlmn** (extend both the group to the left of the **X** and the group to the right of the **X**) or **ijklm** (make the "bad five-letter successor group" into a "good five-letter successor group"); *c.* **ijklpqrs**; *d.* **nmlkji** or **mlkjih**; *e.* **iiiii** (not **iiiij**; here, "extend the successor group" is translated into "extend the repetitive group"); *f.* **iiiijjjjkkkk**. (The problems in this family illustrate how even as seemingly simple a concept as "extend a group" is blurry and can bend in different ways under different pressures.)

5. *a.* **rlyg**; *b.* **ljsrezv** or **ooooo** (the rule is either "extract all isolated letters" or "extract the 'black sheep'").

6. *a.* **xxx**; *b.* **efghij**; *c.* **wxyz**; *d.* **pqrs** (better than **ponm**); *e. 1.* **cd**; *2.* **cde** or **cba**.

7. *a.* **spsss**; *b.* **osooo**; *c. 1.* **dcbabcd**; *2.* **abbbc** or **bbbacbbb** (exploring the concept of "turning a string inside out").

8. *a. 1.* **qjk**; *2.* **qyz** or **xyq**; *b.* **xya** (this problem emphasizes the symmetry between concepts **A** and **Z**).

9. *a.* **pqrstu**; *b.* **pqrs** or **ppqqrrss** (these problems illustrate the notion of "fixing up" a string).

10. *a.* **y**; *b.* **a** (better than **z**); *c.* **z** (better than **a**). (The answers to these last two problems come from a very abstract resemblance between two pairs of symmetric- opposites: the pair **pqr**, **rqp** and the pair **a**, **z**.)

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